Adversarial Transformer-Based Anomaly Detection for Multivariate Time Series

Xinying Yu, Kejun Zhang, Yaqi Liu, Bing Zou, Jun Wang, Wenbin Wang, and Rong Qian

***Abstract*—Anomaly detection in multivariate time series is crucial to monitor system status, such as fault detection in industrial systems. However, detecting anomalies in multivariate time series is challenging due to few labels, complex spatiotemporal correlations, and ultrafast detecting demands. Existing anomaly detection methods rarely address these challenges simultaneously. Herein, we design an adversarial transformers-based unsupervised anomaly detection model (ATUAD). In ATUAD, a Transformer-based encoder–decoder is constructed to learn sequence features, and adversarial training is adopted to amplify mild anomalies and enhance the robustness. Besides, we propose a peak-over-threshold-based dynamic threshold mechanism to improve the anomaly detection performance of ATUAD by automatically determining the threshold. In addition, we provide an anomaly explanation method to help ATUAD pinpoint root causes for anomalies. Comparison experiments, ablation studies, and overhead analysis on public datasets show that ATUAD can outperform the state-of-the-art baseline methods.**

***Index Terms*—Adversarial training, anomaly detection, multivariate time series, self-attention, transformer.**

I. Introduction

THE proliferation of Internet of Things (IoT) devices leads to a large amount of real-time data with more complexity generated across cyber-physical systems [1]. This kind of data consists of several variables (or features), which change over time and are known as multivariate time-series data. IoT applications, such as smart factories or supply chain systems [2], typically adopt multivariate time series to monitor system status. However, harsh natural environments, sensors' hardware problems, or network attacks may result in anomalous data. A general definition of an anomaly within the context of IoT is the measurable consequences of an unexpected change in the state of a system that is outside of its local or global norm [3]. The anomalous data can affect the correctness of IoT application decisions, reduce the quality of IoT services, and cause economic losses. Therefore, it becomes a material issue to detect anomalous data for core tasks in IoT.

Anomaly detection usually involves the identification of novel or unexpected observations or sequences within the captured data. Unfortunately, anomaly detection in multivariate time-series data is more arduous because of the complex spatiotemporal correlations and ultrafast detection demands. Moreover, there is no shortage of mild anomalies close to normality in multivariate time series with few or no labels, which complicates anomaly detection. Therefore, it is paramount to investigate unsupervised anomaly detection for multivariate time series.

To detect anomalies in multivariate time series, numerous deep learning-based unsupervised anomaly detection algorithms have been proposed [4]. The most representative deep learning methods include convolutional neural networks (CNNs) [5], [6], recurrent neural networks (RNNs) [7], and generative adversarial networks (GANs) [8], [9]. Researchers often subtly combine CNNs and RNNs to construct more effective anomaly detection models. However, recurrent models, such as LSTMs, are known to be computationally expensive, and CNNs fail to capture long-distance features, which are unsuitable for environments requiring high computational efficiency (e.g., IoT) [10]. GAN-based anomaly detection methods train both generators and discriminators in an adversarial manner and use the trained discriminators as anomaly detectors. Regrettably, GAN is prone to problems, such as pattern collapse and nonconvergence during the training process [11].

Recent studies have shown the powerful capacity of Transformers to enhance sequence modeling and decrease computational overheads [12]. The Transformer is a network architecture eschewing recurrence and convolutions and instead depending solely on attention mechanisms [13]. Besides, the Transformer can process long sequences in parallel, and its accuracy and training time are almost unaffected by the sequence length [14]. However, using only the simple Transformer-based encoder–decoder model leans toward neglecting anomalies if they are relatively verge on normality. The adversarial training of GAN can magnify the reconstruction error of anomalous data, helping the Transformer model to identify mild anomalies while gaining stability. Thus, we jointly consider Transformer and adversarial training to implement anomaly detection and diagnosis for multivariate time series.

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In this article, an adversarial transformer-based unsupervised anomaly detection model (ATUAD) is developed to deal with the complex spatiotemporal correlations and mild anomalies undetected issues in multivariate time series and fast detection demands in modern applications. Specifically, we construct a Transformer-based encoder–decoder architecture associated with an adversarial training framework to capture long-distance contextual information while alleviating the omission of mild anomalies. Further, we propose a peak-over-threshold (POT) method-based dynamic threshold (PDT) mechanism and introduce it into ATUAD to determine the threshold for detecting anomalies. Then, we conduct an anomaly explanation on the detected anomalous windows to help pinpoint anomaly root causes. By performing extensive experiments, we corroborate that ATUAD outperforms most baselines under diversified metrics. The main contributions are as follows.

1) A novel ATUAD is proposed for multivariate time series. Specifically, in our ATUAD, a shared encoder-different decoder branch is elaborately designed in a transformer style to form two transformer blocks. ATUAD executes reconstruction training by minimizing the reconstruction errors.
2) An adversarial training is designed by integrating with our carefully devised transformer blocks. ATUAD executes adversarial training across two transformer blocks, and one block maximizes reconstruction errors while the other minimizes ones. The adversely trained transformer networks oblige ATUAD to improve detecting accuracy while gaining stability.
3) A PDT is proposed to choose the threshold. PDT improves the streaming POT (SPOT) and learns a global threshold from the anomaly scores to detect anomalous windows, which supplies the benefit of being able to update the threshold automatically. Further, an anomaly explanation method is provided to help ATUAD pinpoint anomaly root causes. Extensive experiments on public datasets verify the effectiveness of ATUAD in detecting performance and training efficiency.

The rest of this article is organized as follows. Section II overviews the works related to the anomaly detection of multivariate time series. Section III constructs the ATUAD and elaborates on the process of anomaly inference. Section IV analyzes the experimental results of ATUAD and the baseline methods. Finally, Section V concludes this article.

II. Related Work

Deep learning methods have a strong ability to handle nonlinearity in temporal correlation and have achieved remarkable results in many time-series anomaly detection methods.

In pursuit of higher detection accuracy, many time series anomaly detection efforts have applied recurrent neural networks. Zhang et al. [15] proposed MSCRED, which employs a multiscale convolutional encoder–decoder and a temporal attention-based convLSTM for anomaly detection and diagnosis in multivariate time series. MSCRED achieves anomaly detecting, anomaly root identifying, and anomaly severity reflecting, which effectively help system operators perform timely system diagnosis and repair. However, the model uses a given anomaly threshold for detection, which makes the model less applicable and less general. Su et al. [16] proposed OmniAnomaly, a stochastic recurrent neural network for multivariate time series anomaly detection. OmniAnomaly learns a robust representation of data using techniques, such as stochastic variable connection and planar normalizing flow, and identifies anomalies based on reconstruction probabilities. The method achieves remarkable anomaly detection performance and anomaly interpretation accuracy, but at the cost of high training time. MAD-GAN [17] is an unsupervised multivariate anomaly detection method based on generative adversarial networks. The generators and discriminators of GANs use LSTM to capture the temporal correlation in time series distributions. However, the model is computationally expensive and cannot effectively model large sequences.

Subsequently, research scholars have increasingly emphasized the inference efficiency of anomaly detection methods. Zong et al. [18] proposed DAGMM, an unsupervised anomaly detection model combining a deep autoencoder and a Gaussian mixture model. The deep autoencoder compresses the data points into a low-dimensional representation and inputs the Gaussian mixture model to evaluate the density of the representation. However, the model targets multivariate variables rather than multivariate time series, ignoring the inherent time dependence of the time series. Audibert et al. [11] proposed USAD, a multivariate time series unsupervised anomaly detection method based on autoencoder and adversarial training. USAD processes sequence data through an autoencoder with two decoders and an adversarial training framework, and determines anomalies based on reconstruction errors. Compared with prior art, USAD can significantly reduce training time. However, both the encoder and decoder in the model use fully connected neural networks, which sometimes cannot extract sequence features effectively.

The transformer architecture is favored by more and more researchers, which provides a new direction for time series anomaly detection. Chen et al. [10] proposed GTA, a Transformer-based framework for anomaly detection that uses the introduced connection learning policy to automatically learn sensor dependencies. Also, an influence propagation graph convolution is applied to simulate the information flow among the sensors in the graph. They claimed that the inference speed of the multibranch attention technique is improved without sacrificing model performance. Tuli et al. [14] proposed Transformer-based anomaly detection and diagnosis (TranAD). TranAD uses the attention-based Transformer encoder–decoder framework, focuses score-based self-adjustment, and adversarial training to reconstruct time-series data, achieving superior detection performance and training efficiency on multiple datasets. Xu et al. [19] proposed TGAN-AD, a transformer-based GAN method for anomaly detection, whose generator aids in extracting contextual features of time-series data, and discriminator assists in determining abnormal data. They measure the anomalies by calculating both the reconstruction loss of the generator and the discrimination loss of the discriminator. Li et al. [20] proposed a dilated convolutional transformer-based GAN (DCT-GAN) to solve the problems of time series anomaly detection.

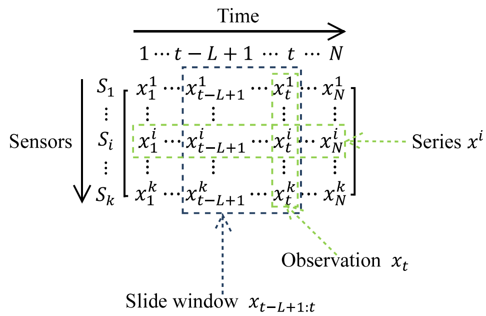


Fig. 1. Data formulation of the multivariate time series.

DCT-GAN utilizes several generators and a single discriminator to alleviate the mode collapse problem. Each generator consists of a dilated convolutional neural network and a Transformer block to obtain fine-grained and coarse-grained information about the time series. The experiments on the NAB dataset show that DCT-GAN is more accurate and stable than most other methods. Despite their effectiveness, they are few to consider the temporal dependency, mild abnormalities, and inference efficiency, simultaneously. Therefore, we propose ATUAD to cope with the above problems and improve the precision and efficiency of anomaly detection with low computational cost.

III. ATUAD Model

A. Problem Statement

In this article, the training data (i.e., multivariate time series) $X = \{x_1, x_2, \ldots, x_N\} \in R^{k\times N}$ are composed of the measurement data of k sensors over N consecutive timestamps. As shown in Fig. 1, the observation $x_t \in R^k$ at any timestamp t is a k-dimensional vector representing the measurement data of k sensors, and x_t^i denotes the value of the ith sensor S_i at the timestamp t. Particularly, the univariate time series $X \in R^{k\times N}$ $(k = 1)$ is a special case in the multivariate setting, where x_t is a scalar. Similar to previous unsupervised anomaly detection works, this article only models sequence on normal data (without anomalies) and detects anomalies on test data (with anomalies). To model the dependence between the observation x_t and the historical ones, we consider a slide time window $W_t = \{x_{t-L+1}, \ldots, x_t\}$ of length L, both for training and testing. Instead of using the original multivariate time series X, a sequence of windows $\mathcal{W} = \{W_1, \ldots, W_T\}$ serves as the model's input. We utilize the replication padding [19] to maintain each window length as L, i.e., $W_t = \{x_1, \ldots, x_t, \ldots, x_t\}$ when $t < L$. Our model takes as inputs $\mathcal{W}_{\text{train}} = \{W_1, \ldots, W_T\}$ for training and $\hat{\mathcal{W}}_{\text{test}} = \{\hat{W}_1, \ldots, \hat{W}_{\hat{T}}\}$ for testing and assigns a binary label for each testing window $\hat{W}_t$ to indicate an anomaly or not based on the global threshold and the window's anomaly score. We now formalize the two problems of anomaly detection and diagnosis.

Anomaly detection: Given testing data $\hat{\mathcal{W}}_{\text{test}} = \{\hat{W}_1, \ldots, \hat{W}_{\hat{T}}\}$, predict binary labels $y = \{y_1, \ldots, y_{\hat{T}}\}$, where $y_t \in \{0, 1\}$ indicates whether the window $\hat{W}_t$ at timestamp t is anomalous $(y_t = 1)$ or not $(y_t = 0)$.

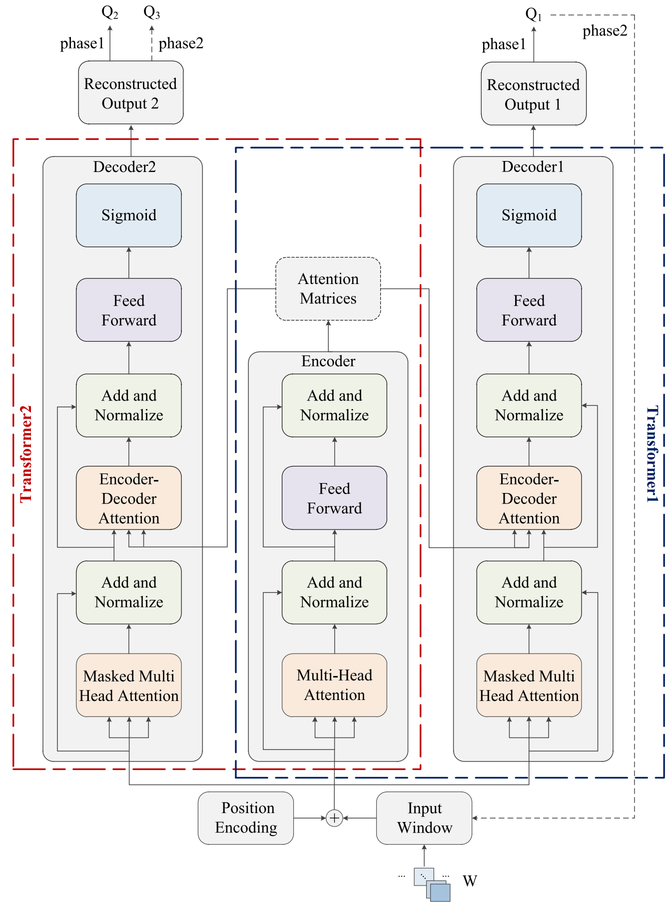


Fig. 2. Overall architecture of ATUAD. In the visualization of ATUAD's architecture with one encoder layer and two decoder layers, three components form two transformer blocks by sharing the same Encoder. The encoder encodes the concatenation of the input window and the position encoding and projects it to a latent variable, which then undergoes two decoders to generate reconstructions. The two transformer blocks are trained in an adversarial way, where Transformer1 seeks to fool Transformer2, and Transformer2 aims to distinguish the input reconstructed by Transformer1 from the real data.

Anomaly diagnosis: Given the above detection results $y = \{y_1, \ldots, y_{\hat{T}}\}$, recognize the sensors most likely responsible for the anomalies within the anomalous window $\hat{W}_t$.

B. Overall Architecture of ATUAD

Transformer is a popular deep learning model that has been widely used in natural language processing and sequence modeling tasks due to the superior capability of multihead attention mechanism in long-distance dependencies capturing. Fig. 2 shows the architecture of the neural network used in our model. For simplicity, we use E, D_1, D_2, and W to symbolize the Encoder, Decoder1, Decoder2, and W_t, separately. The model's specifics are implemented below.

ATUAD design: The classical transformer-based encoder–decoder model is designed to excel at reconstructing normal data while failing to do so for anomalous data that deviates from the learned distribution, thus detecting anomalous data based on the noticeable deviations. However, if the anomalous data are relatively close to normality, the transformer-based encoder–decoder model tends to miss the anomalous data. Therefore, the model should be able to recognize whether the input data does not contain anomalies before conducting a good reconstruction. Given that GAN models perform well in the characterization tasks of determining whether the input data is abnormal, we introduce a GAN-like style adversarial training method in the transformer-based encoder–decoder model to detect anomalies.

As portrayed in Fig. 2, we concatenate position encoding with a training window to obtain the input I for the transformer encoder, which projects I into an attention matrix Z. Next, the two transformer decoders reconstruct the input using the attention matrix separately. The specific operations of the transformer encoder are as follows:

$$I = W \oplus PE \tag{1}$$

$$I_1 = \text{LayerNorm}(I + \text{MultiHeadAtt}(I, I, I)) \tag{2}$$

$$Z = \text{LayerNorm}(I_1 + \text{Feedforword}(I_1)) \tag{3}$$

where PE, $\oplus$, and $+$ denote position encoding, concatenation operation, and matrix addition, separately. MultiHeadAtt() denotes the multihead self-attention mechanism, Feedforword() represents the fully connected feedforward network, and LayerNorm() denotes the residual connection and layer normalization operations. The transformer encoder generates attention weights by processing all dimensions of the time window in parallel to capture the temporal trends in the input sequence, which significantly improves the model's training time. The attention matrix output from the encoder is used as the key and value of the encoder–decoder attention layer in transformer decoders, thus assisting the decoders to focus on the sequence's appropriate position. The specific operations of the transformer decoders are as follows:

$$I_2 = \text{Mask}(\text{MultiHeadAtt}(I, I, I)) \tag{4}$$

$$I_3 = \text{LayerNorm}(I_2 + I) \tag{5}$$

$$I_4 = \text{LayerNorm}(I_3 + \text{EncDecAtt}(I_3, Z, Z)) \tag{6}$$

$$Q_i = \text{Sigmoid}(\text{Feedforword}(I_4)). \tag{7}$$

For time-series data, the decoder's output at timestamp t should depend only on the outputs before t, but not on the data at future timestamps. Therefore, the multihead attention mechanism in the decoders uses a sequence mask to hide the data at subsequent positions. Sigmoid function is used to normalize the reconstructions to match the preprocessed input window. Eventually, the ATUAD model generates two reconstruction windows Q_i ($i \in \{1, 2\}$) using the input window W.

C. Two-Phase Adversarial Training

The model's training consists of two phases: the reconstruction training and the adversarial training. First, the two transformer blocks are trained to learn to reconstruct the normal data as much as possible. Second, the two transformer blocks are trained in an adversarial way. Specifically, Transformer1 (Encoder+Decoder1) refeeds the reconstruction Q_1 from phase 1 into the model, attempting to deceive Transformer2 (Encoder+Decoder2), while Transformer2 learns to distinguish whether the data is from the input window W or Transformer1's reconstruction.

Phase 1—Reconstruction training: The intent of phase 1 is to generate an approximate reconstruction of the input window by training two transformer blocks. The reconstruction error of each block is defined as the L_2 norm of input window W and reconstruction Q_i

$$L_1 = \|W - Q_1\|_2 \tag{8}$$

$$L_2 = \|W - Q_2\|_2. \tag{9}$$

Phase 2—Adversarial training: In the second phase, the transformer encoder recompresses the reconstruction generated by Transformer1, and then the recompressed result will be decoded again by Decoder2. In this phase, the objective of Transformer1 is to fool Transformer2 by minimizing the deviation between the input window W and the reconstruction Q_3. The objective of Transformer2 is to distinguish W from Q_1 by maximizing the difference between W and Q_3. It is worth noting that Transformer2 of our model does not strictly play the role of the discriminator of GAN. The output of Transformer2 in Phase 2 is not a scalar but a reconstruction of the result generated from Transformer1 in Phase 1. The training goals of the model in phase 2 are as follows:

$$L_1 = +\|W - Q_3\|_2 \tag{10}$$

$$L_2 = -\|W - Q_3\|_2. \tag{11}$$

Two phases training: The objective function of each transformer block contains two components, the reconstruction loss and the adversarial loss. Transformer1 minimizes the reconstruction error between the input window W and the reconstruction Q_1 and minimizes the adversarial error between W and Q_3. Similar to Transformer1, Transformer2 minimizes the reconstruction error between the input window W and the reconstruction Q_2, while maximizing the adversarial error between W and Q_3. Unlike GANs training the discriminator while fixing the generator, we use an evolutionary scheme for each transformer block that combines the reconstruction loss and the adversarial loss to train two blocks simultaneously

$$L_1 = \frac{1}{n}\|W - Q_1\|_2 + \left(1 - \frac{1}{n}\right)\|W - Q_3\|_2 \tag{12}$$

$$L_2 = \frac{1}{n}\|W - Q_2\|_2 - \left(1 - \frac{1}{n}\right)\|W - Q_3\|_2 \tag{13}$$

where n denotes the training epoch. Algorithm 1 summarizes the complete training process of the ATUAD model. The proportion of reconstruction loss and adversarial loss in the objective function varies with the training epoch. Initially, the weight of the reconstruction loss is high, which ensures stable training when the outputs of the decoders are poor reconstructions of the input windows. With poor reconstructions, the model training of the second phase would be unreliable. To this end, the weight of the adversarial loss is low in the inception to avoid destabilizing model training. As the reconstructions approach the input windows, the adversarial loss weight increases gradually. The

Algorithm 1: Training Algorithm of ATUAD.

```
Input: Training epochs N; Sequence of training
  windows W_train = {W_1, ..., W_T}
Output: The trained model ATUAD
  E, D_1, D_2 ← Initialize weights
  n ← 1
  do
    for t = 1 to T do
      Z_t ← E(W_t); Q_1 ← D_1(Z_t)
      Q_2 ← D_2(Z_t); Q_3 ← D_2(E(Q_1))
      L_1 ← (1/n)||W_t − Q_1||_2 + (1 − 1/n)||W_t − Q_3||_2
      L_2 ← (1/n)||W_t − Q_2||_2 − (1 − 1/n)||W_t − Q_3||_2
      E, D_1, D_2 ← Update weights using L_1, L_2
    end for
    n ← n + 1
  while n < N
```

Algorithm 2: Peaks-Over-Threshold (POT).

```
procedure POT(X_1, ..., X_n; q)
  θ ← SetInitialThreshold(X_1, ..., X_n)
  Y_θ ← {X_i − θ | X_i > θ} // Y_θ is a set of peaks
  γ̂, σ̂ ← Grimshaw(Y_θ)
  z_q ← CalcThreshold(q, γ̂, σ̂, n, N_θ, θ)
  return z_q, θ
end procedure
```

Algorithm 3: POT-Based Dynamic Threshold Mechanism.

```
procedure PDT(X_1, ..., X_n, X_{n+1}, ..., X_m; q)
  Y_θ ← ∅; TH ← ∅
  z_q, θ ← POT(X_1, ..., X_n; q)
  u ← n
  for n < i ≤ m do
    if X_i > θ then
      Y_i ← X_i − θ
      Add Y_i in Y_θ
      N_θ ← N_θ + 1
      u ← u + 1
      γ̂, σ̂ ← Grimshaw(Y_θ)
      z_q ← CalcThreshold(q, γ̂, σ̂, u, N_θ, θ)
    else
      u ← u + 1
    end if
    Add z_q in TH
  end for
  Th ← CalcAverage(TH)
  return Th
end procedure
```

adversarial training of the transformer-based encoder–decoder architecture enables ATUAD to learn how to magnify the reconstruction error of input windows with anomalies and tend to achieve stability compared to GAN architectures, helping to reduce the risks of overfitting and nonconvergence.

D. POT-Based Dynamic Threshold Mechanism

Extreme value theory (EVT) is a statistical theory whose objective is to discover the laws of extreme events (e.g., the law of daily temperature maxima) [21]. Such laws of extreme events are also called Extreme Value Distributions. For most distributions, the probability decreases as the event approaches an extreme value, i.e., $P(X > x) \to 0$ as x increases. The function $\overline{F}(x) = P(X > x)$ designates the "tail" of the probability distribution of X. It is possible to evaluate the probability of a potential extreme event by fitting the extreme value distribution to the tail of the input distribution. In particular, given a probability q, we can calculate a threshold z_q such that $P(X > z_q) < q$.

One way to fit the tail of the distribution is the peak-over-threshold (POT) approach, also called the second theorem in EVT [21]. The basic idea of POT is to fit the tail of the probability distribution using the generalized Pareto distribution (GPD), as shown in the following:

$$\overline{F}_\theta(x) = P(X - \theta > x | X > \theta) \sim \left(1 + \frac{\gamma x}{\sigma(\theta)}\right)^{-\frac{1}{\gamma}}. \tag{14}$$

Equation (14) shows that the excess over an initial threshold θ, written as $X - \theta$, satisfies the GPD with a shape parameter γ and a scale parameter σ. After obtaining estimates $\hat{\gamma}$ and $\hat{\sigma}$ by maximum likelihood estimate and Grimshaw method [22], the threshold z_q can be calculated through the following equation:

$$z_q \simeq \theta + \frac{\hat{\sigma}}{\hat{\gamma}}\left(\left(\frac{qn}{N_\theta}\right)^{-\hat{\gamma}} - 1\right) \tag{15}$$

where the risk value q is the given probability. n denotes the number of observations $X_1, \ldots, X_n$. N_θ is the number of peaks $X_i - \theta$, where $X_i > \theta$. Algorithm 2 summarizes the POT approach to determine the threshold z_q.

SPOT [21] is a streaming anomaly detection method that applies the POT model and dynamic thresholds to detect anomalies in streaming data $(X_i)_{i>0}$. The threshold updates of SPOT only deal with the peak cases without considering the anomaly cases. The small peak set will make the model more variable (high variance). Inspired by the dynamic updating of thresholds in SPOT, we develop a PDT, which omits the anomaly detection process in SPOT and brings the anomaly cases into the scope of threshold updates to obtain a precise threshold that could effectively distinguish the actual anomalies from benign samples. Initially, we conduct the POT approach on the first n observations to reach initial thresholds z_q and θ. Then, we update z_q based on the magnitude between the next observed values and θ. If a value exceeds θ, we will include the corresponding peak in the peak set and update z_q. Finally, we add all the thresholds z_q to the threshold set and treat the mean as the global threshold. Algorithm 3 describes the PDT mechanism.

E. Anomaly Inference

Anomaly detection: The preprocessed test data are fed into the trained ATUAD model for anomaly detection. The anomaly detection process also executes two phases and finally obtains a pair of outputs $(\hat{Q}_1, \hat{Q}_3)$. As a new testing window (say $\hat{W}_t$) arrives, the model calculates its anomaly score as follows:

$$\hat{s}_t = \frac{1}{2}\|\hat{W}_t - \hat{Q}_1\|_2 + \frac{1}{2}\|\hat{W}_t - \hat{Q}_3\|_2 \tag{16}$$

where $\hat{s}_t \in R^{L\times k}$ contains the anomaly score of each dimension at each timestamp in the test window. We use the proposed PDT mechanism to learn the global threshold Th of the anomaly scores

$$\mathrm{Th} = \mathrm{PDT}(s_1, \ldots, s_T, \hat{s}_1, \ldots, \hat{s}_{\hat{T}}; q) \tag{17}$$

where $\{s_1, \ldots, s_T\}$ is the set of anomaly scores of the training windows, which is used in the POT step in Algorithm 3 to obtain the initial threshold z_q. $\hat{s}_1, \ldots, \hat{s}_{\hat{T}}$ is the set of anomaly scores of the testing windows, which is used to obtain the global threshold Th in Algorithm 3. Since the POT algorithm requires the first input to be a univariate sequence, we transform s_t $(t = 1, \ldots, T)$ or $\hat{s}_t$ $(t = 1, \ldots, \hat{T})$ into a scalar representing the mean of the anomaly scores of the tth time window. If the score of the test window $\hat{W}_t$ is higher than the global threshold, the window will be declared as anomalous, i.e., $y_t = 1$.

Anomaly diagnosis: To pinpoint the specific sensors leading to anomalies, we calculate the residual matrix of the abnormal testing window and its reconstruction window

$$\Delta\hat{W} = \|\hat{W}_t - \hat{Q}_3\|. \tag{18}$$

For each dimension $\Delta\hat{W}^i = \{\Delta\hat{W}^i_1, \ldots, \Delta\hat{W}^i_L\}$ $(\Delta\hat{W}^i \in R^L, i = 1, \ldots, k)$ in the residual matrix, the local threshold is calculated by using the POT algorithm

$$\mathrm{th}_i = \mathrm{POT}(\Delta\hat{W}^i, q). \tag{19}$$

It indicates that the sensor i has caused abnormal data at timestamp l if there is a difference $\Delta\hat{W}^i_l$ in $\Delta\hat{W}^i$ higher than the local threshold th_i. A longer sequence of consecutive scores in $\Delta\hat{W}^i$ higher than the local threshold indicates a longer anomaly duration and a higher anomaly severity level. Algorithm 4 summarizes the anomaly detection and diagnosis process of the ATUAD model.

Algorithm 4: Detection Algorithm of ATUAD.

```
Input: The trained model; risk value q; Sequence of
  training windows W_train = {W_1, ..., W_T}; Sequence
  of testing windows Ŵ_test = {Ŵ_1, ..., Ŵ_T̂}
Output: Labels y = {y_1, ..., y_T̂}
  for t = 1 to T do
    Q_1 ← D_1(E(W_t)); Q_3 ← D_2(E(Q_1))
    s_t ← 1/2 ||W_t − Q_1||_2 + 1/2 ||W_t − Q_3||_2
  end for
  for t = 1 to T̂ do
    Q̂_1 ← D_1(E(Ŵ_t)); Q̂_3 ← D_2(E(Q̂_1))
    ŝ_t ← 1/2 ||Ŵ_t − Q̂_1||_2 + 1/2 ||Ŵ_t − Q̂_3||_2
  end for
  Th ← PDT(s_1, ..., s_T, ŝ_1, ..., ŝ_T̂; q)
  for t = 1 to T̂ do
    if ŝ_t > Th then
      y_t ← 1; ΔŴ ← ||Ŵ_t − Q̂_3||
      for i = 1 to k do
        th_i ← POT(ΔŴ^i, q)
        if ΔŴ^i_l > th_i then
          sensor i is abnormal at time l
      end for
    else
      y_t ← 0
  end for
  return y = {y_1, ..., y_T̂}
```

TABLE I
DATASET INFORMATION

Dataset	Train	Test	Dimensions	Anomalies (%)
MBA	100 000	100 000	2	0.14
SMAP	135 183	427 617	25	13.13
MSL	58 317	73 729	55	10.72
SWaT	496 800	449 919	51	11.98
WADI	1 048 571	172 801	123	5.99
SMD	708 405	708 420	38	4.16

IV. EXPERIMENTS

A. Experimental Setup

Datasets: The datasets adopted in the experiment include MIT-BIH Supraventricular Arrhythmia Dataset (MBA) [24], Soil Moisture Active Passive Dataset (SMAP) [25], Secure Water Treatment Dataset (SWaT) [27], Water Distribution Dataset (WADI) [28], Mars Science Laboratory Dataset (MSL) [26], and Server Machine Dataset (SMD) [16], which are commonly used in multivariate anomaly detection. I summarizes the characteristics of each dataset. We abandon Yahoo and Numenta datasets due to the flaws, such as mislabeled ground truth and run-to-failure bias reported in [23].

Training setting: We train all models employing PyTorch-1.8.0 on a Ubuntu 18.04 system with configuration RTX 4090 GPU. Our model has one transformer encoder layer and two transformer decoder layers. For time series reconstructing, we set the window size to ten frames with a batch size of 128. The number of heads in multihead attention is consistent with the dimensionality of the dataset. We utilize the dropout strategy to prevent overfitting with a dropout rate of 0.1. In addition, we train the models using the AdamW optimizer with a corresponding learning rate for each dataset. Compared with Adam, the weight decay of AdamW achieves decoupling and better results. Table II shows the hyperparameters used in our ATUAD model. Table III gives the hyperparameters of baselines used in our experiment.

B. Performance Analysis

Anomaly detection results comparing with baselines: To demonstrate the anomaly detection performance, we compared our model with six state-of-the-art methods, namely, MSCRED [15], OmniAnomaly [16], DAGMM [18], MAD-GAN [17], USAD [11], and TranAD [14]. Table IV shows the results of all models on six datasets. From Table IV, we can conclude that, except for the SWaT dataset, our ATUAD achieves

TABLE II
HYPERPARAMETERS OF OUR MODEL

Parameters	Values
Window length	10
Batch size	128
Epochs	5
Dropout	0.1
Risk value q	0.00001
Number of Encoder (Decoder)	1 (2)
Input size of Encoder (Decoder)	$*$
Hidden size of Encoder (Decoder)	16
Output size of Encoder (Decoder)	$*$
Learning rate	0.002(MSL) 0.008(SWaT) 0.01(SMAP, MBA) 0.0001(SMD, WADI)

$*$: Related to the dataset's dimensions

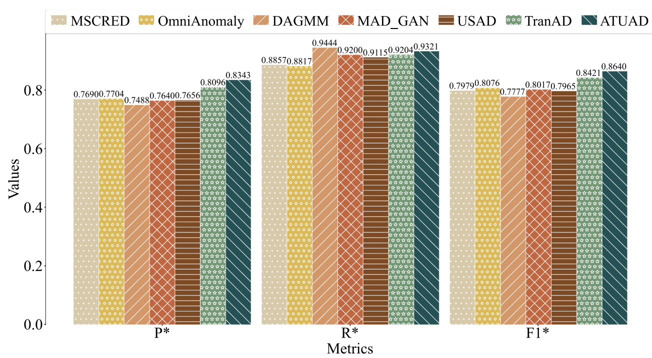


Fig. 3. Average performance of all models.

better F1 scores than other methods across all datasets. USAD realizes the best F1 score on the SWaT dataset, and the F1 score of our model is close to that of USAD. Furthermore, the performance of our ATUAD is improved on most datasets, especially on the WADI dataset, with an improvement of approximately 9% and 10% in P and $F1$ compared to the optimal baseline, respectively. Fig. 3 portrays the average performance of all models on six datasets. As shown in Fig. 3, the average precision (P^*) and the average F1 score ($F1^*$) of our ATUAD exceed those of other models. Specifically, our ATUAD achieves an improvement of up to 2.47% in P^* and 2.19% in $F1^*$ compared to the optimal baseline, but the average recall (R^*) is 1.23% lower than that of DAGMM.

DAGMM [18] reaches high performance on short datasets like MBA and SMAP. However, DAGMM performs poorly on large sequence datasets, such as WADI, since it ignores the inherent temporal dependence in multivariate time series. The temporal dependence is paramount for the time-series data since the observations are dependent. Therefore, the temporal information in historical data is essential for reconstructing or predicting the time series. In our ATUAD, the multivariate time-series data are transformed into a sequence of windows to hold the temporal information. Furthermore, ATUAD uses transformer blocks as the feature extractor to model the serial data. Consequently, ATUAD performs well on the longer sequence of high-dimensional datasets.

MSCRED [15] and OmniAnomaly [16] take as input the observation sequences and consider LSTM or GRU recurrent neural networks to capture the temporal information in the sequential observations, achieving high F1 scores on MBA and MSL. However, such methods identify anomalies merely by reconstructing, thus failing to detect slight outliers closer to normal data in datasets like SMD. Our ATUAD utilizes adversarial training to boost the deviation and has a high detection sensitivity for anomalous data with mild anomalies.

MAD-GAN [17] adopts adversarial training to reconstruct the original time series, so it performs better than MSCRED on the SMD dataset with mild anomalies. Nevertheless, MAD-GAN also uses RNNs as the basic framework, which restricts the modeling capability over the long-term sequence. Our ATUAD applies the powerful multihead attention mechanism to learn the sequence features in a parallel fashion, achieving swift calculation efficiency and the capability of capturing long-distance context information.

USAD [11] uses an autoencoder framework with an adversarial style to achieve 94.95% and 94.47% F1 scores on SMD and MBA, respectively, but performs mediocrely on other datasets, especially on the high-dimensional and unbalanced datasets, such as WADI. This is because USAD simply uses linear transformations to encode and decode the data, which speeds up the training but neglects the temporal correlation within the multivariate time-series data.

TranAD [14] extracts multimodal features leveraging focus score-based self-conditioning and gains stability using adversarial training. It outperforms other models on SWaT and SMD. However, the F1 score of TranAD on the WADI dataset is 49.51%, which is lower than the F1 score of 59.16% achieved by our ATUAD. This improvement can be attributed to the input window and adversarial training. The input windows fed into ATUAD contain richer temporal information than that of TranAD, facilitating the learning of time series features. Specifically, ATUAD takes as input the data of a complete batch of time windows to its decoders for aiding temporal attention, while just one local contextual window is fed into the TranAD's Window Encoder. In addition, ATUAD conducts adversarial training using the reconstruction of the input directly instead of a sparse focus score matrix used in TranAD. Therefore, our ATUAD has an advantage in obtaining the temporal trend of long-term sequences and achieves better results even on high-dimensional and unbalanced datasets.

Root cause identification results: To further analyze the effectiveness of ATUAD, Fig. 4 presents a case study of anomaly diagnosis in the SMD dataset. In Fig. 4(a), the red dashed line represents the global threshold, and the areas highlighted by blue rectangles portray the anomalous fragments in SMD. ATUAD declares the sequences with anomaly scores exceeding the threshold as abnormal sequences. As depicted in Fig. 4(a), multitudinous abnormal sequences in the SMD testing dataset have scores higher than the threshold, which indicates that ATUAD can pinpoint the actual positives with false positives that are not excessive.

The purple matrix in Fig. 4(b) represents the residual matrix between the input sequence and the reconstructed sequence.

TABLE III
HYPERPARAMETERS OF BASELINES

Baselines	Learning rate	Window length	Encoder/Generator				Decoder/Discriminator			
			Input size	Hidden size	Output size	Linear/Conv layers	Input size	Hidden size	Output size	Linear/Conv layers
MSCRED [20]	0.0001	∗	1	32, 64	128	3	128	64, 32	1	3
OmniAnomaly [21]	0.002	1	32	32	16	3	8	32	∗	3
MAD-GAN [22]	0.0001	5	∗	16	∗	3	∗	16	1	3
DAGMM [23]	0.0001	5	∗	16	8	3	8	16	∗	3
USAD [16]	0.0001	5	∗	16	5	3	5	16	∗	3
TranAD [19]	0.0001	10	∗	16	∗	2	∗	16	∗	2

∗: Related to the dataset's dimensions

TABLE IV
PERFORMANCE COMPARISON OF ALL MODELS

Methods		SMAP			MSL			SWaT		
		P	R	F1	P	R	F1	P	R	F1
CNN and RNN-based	MSCRED [20]	0.8175	0.9216	0.8664	0.8912	0.9862	0.9363	0.9992	0.6770	0.8071
VAE and RNN-based	OmniAnomaly [21]	0.7974	0.9419	0.8636	0.7848	0.9999	0.8794	0.9782	0.6957	0.8131
GAN and RNN-based	MAD-GAN [22]	0.8122	0.9216	0.8634	0.8516	0.9930	0.9168	0.9593	0.6957	0.8065
AE and FCN-based	DAGMM [23]	0.8069	0.9891	0.8888	0.7209	0.9999	0.8378	0.9933	0.6879	0.8129
	USAD [16]	0.8104	0.9627	0.8800	0.7949	0.9912	0.8822	0.9977	0.6879	**0.8143**
Transformer and FCN-based	TranAD [19]	0.8043	0.9999	0.8915	0.9038	0.9999	0.9494	0.9739	0.6957	0.8116
	ATUAD (ours)	0.8157	0.9999	**0.8985**	0.9041	0.9999	**0.9496**	0.9696	0.6957	0.8101
Methods		WADI			SMD			MBA		
		P	R	F1	P	R	F1	P	R	F1
CNN and RNN-based	MSCRED [20]	0.2513	0.7319	0.3741	0.7276	0.9974	0.8414	0.9272	0.9999	0.9622
VAE and RNN-based	OmniAnomaly [21]	0.3158	0.6541	0.4260	0.8881	0.9985	0.9401	0.8578	0.9999	0.9234
GAN and RNN-based	MAD-GAN [22]	0.2376	0.9124	0.3770	0.7833	0.9974	0.8775	0.9400	0.9999	0.9690
AE and FCN-based	DAGMM [23]	0.1143	0.9981	0.2051	0.9103	0.9914	0.9491	0.9468	0.9999	0.9726
	USAD [16]	0.1891	0.8296	0.3080	0.9060	0.9974	0.9495	0.8953	0.9999	0.9447
Transformer and FCN-based	TranAD [19]	0.3529	0.8296	0.4951	0.8673	0.9974	0.9278	0.9552	0.9999	0.9770
	ATUAD (ours)	0.4407	0.8995	**0.5916**	0.9072	0.9974	**0.9502**	0.9687	0.9999	**0.9841**

P denotes Precision; R denotes Recall; F1 denotes F1 score.

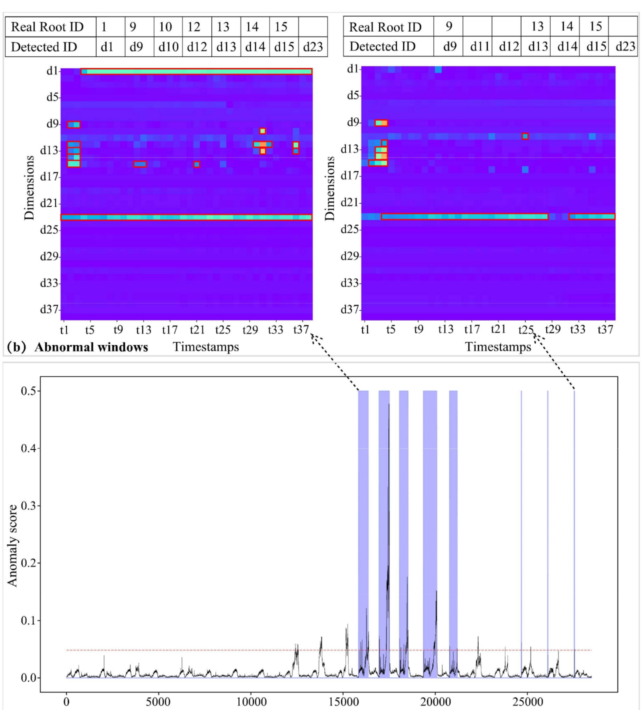


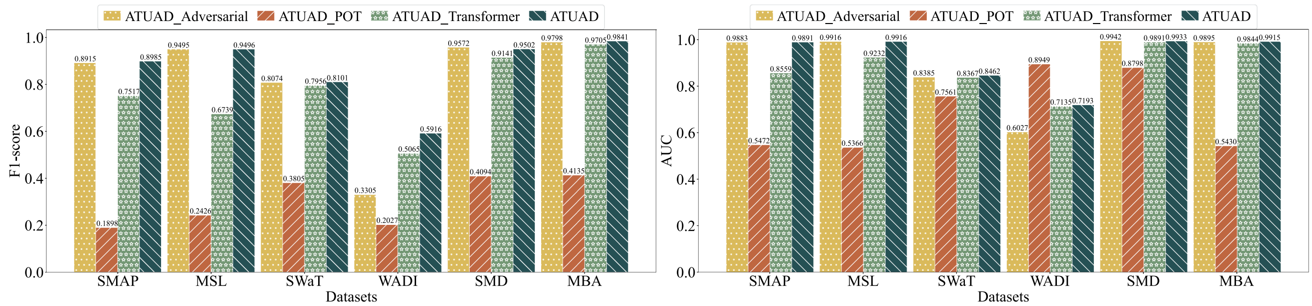
Fig. 4. Case study of anomaly diagnosis. (a) Anomaly score. (b) Abnormal windows.

The red rectangle represents the abnormal events detected by the model. Anomalous events refer to which indicators or dimensions of the detected entity are abnormal in which period. The longer red rectangle represents the longer duration of the abnormality of this indicator, which means higher severity level of the anomaly. In Fig. 4(b), the indicator d1 in the first residual matrix is abnormal at more than 30 consecutive timestamps. It is more likely to occur as a seriously anomalous event. The operator can quickly troubleshoot and fix the root cause of the abnormality based on this information.

In the first anomaly area of the SMD testing dataset, the true anomaly root causes include seven dimensions of 1, 9, 10, and 12–15, while the anomaly root causes detected by ATUAD are d1, d9, d10, d12–d15, and d23 dimensions. In the last anomaly region, the actual anomaly root causes contain four dimensions of 9, 13, 14, and 15, while the anomaly root causes detected by ATUAD are d9, d11, d12, d13, d14, d15, and d23 dimensions. This case indicates that ATUAD can pinpoint 57.10%–87.50% of root causes for anomalies.

C. Ablation Studies

Anomaly detection results comparing with model variants: To investigate the influence of each component on ATUAD, we examine three variants of ATUAD. First, we study ATUAD without adversarial loss. We remove the adversarial training and only consider the reconstruction loss of phase 1, i.e., $L_1 = \|W - Q_1\|_2 + \|W - Q_2\|_2$. Second, to verify the



(a) F1-Score (b) AUC Score

Fig. 5. Performance comparison between ATUAD and variant models. (a) F1-score. (b) AUC score.

TABLE V
COMPARISON OF TRAINING TIME ON ALL DATASETS AND PARAMETER SIZE ON WADI

Method		Training time (s)						Params (MB)
		SMAP	MSL	SWaT	WADI	SMD	MBA	WADI
CNN and RNN-based	MSCRED [20]	16.13	33.47	288.66	1349.05	237.66	592.13	0.35
VAE and RNN-based	OmniAnomaly [21]	24.51	19.06	25.22	216.27	251.05	64.07	0.11
GAN and RNN-based	MAD-GAN [22]	26.09	21.63	27.79	330.94	274.94	62.37	0.12
AE and FCN-based	USAD [16]	19.75	16.22	18.91	**192.93**	208.20	49.37	0.13
	DAGMM [23]	17.06	**13.70**	16.10	**157.58**	**176.38**	41.78	0.12
Transformer and FCN-based	TranAD [19]	**6.42**	**11.20**	**1.17**	275.30	**92.23**	**3.41**	5.16
	ATUAD (ours)	**12.50**	18.75	**1.49**	440.40	181.23	**5.57**	1.35

effectiveness of the POT-based dynamic threshold mechanism, we replace it with the best-F-score threshold selection method [29]. Finally, we substitute the transformer networks with feedforward networks for reconstruction to demonstrate the necessity of a transformer-based structure for modeling sequences. Fig. 5 draws the comparison results of ATUAD and its variants in F1 score and AUC. From Fig. 5, we can report the following conclusions.

ATUAD_Adversarial: After removing the adversarial training, the variant's performance has declined on most datasets, especially the WADI dataset, whose F1 score and AUC have decreased by 26% and 12%, respectively. However, censoring the adversarial training generally has less impact on the performance of the other datasets, and even the SMD dataset exhibits a slight improvement in the F1 score and the AUC. These results reflect that the adversarial training is more suitable for datasets with a large percentage of mild anomalies (e.g., the WADI dataset) by amplifying errors to recognize them.

ATUAD_POT: Replacing the POT-based dynamic threshold mechanism with the best-F-score method resulted in a decrease in the F1 score and AUC for most datasets, which suggests that the anomaly detection model with POT can avoid more underreporting and false alarms on the time-series datasets. The tiniest change is in the WADI dataset, where the F1 score decreased from 59.16% to 20.27%, while the AUC increased slightly. The most significant impact was observed on the SMAP and MSL datasets, with a decrease of approximately 71% and 44% in the F1 score and AUC, respectively. Therefore, the POT-based dynamic threshold mechanism positively influences the model's performance.

ATUAD_Transformer: Substituting the transformer structure with a feedforward network, the variant model has an average decrease of about 10% in the F1-score and 4% in the AUC on all datasets. The decrease is more pronounced on the MSAP and MSL datasets, with a decrease of approximately 15% and 28% in the F1 score and approximately 13% and 7% in the AUC, respectively. These changes demonstrate that the attention-based transformer is more advantageous in identifying anomalies in large-scale datasets.

D. Overhead Analysis

This section investigates the complexity of all models on the given datasets. For this purpose, we measure the average training time in seconds per epoch and model size (Params) and record them in Table V. In general, the three models with the shortest training time are TranAD, DAGMM, and ATUAD, followed by USAD, while the remaining models have longer training time, especially MSCRED. Next, we gradually analyze the reasons for this situation to clarify the effectiveness of our ATUAD. The training times of MSCRED, OmniAnomaly, and MAD-GAN are longer than those of TranAD and ATUAD since the formers use recurrent neural networks for sequential processing while the later apply parallel computing. DAGMM and USAD consist of fully connected neural networks whose training times are generally shorter than those of MSCRED, OmniAnomaly, and MAD-GAN while still higher than TranAD and ATUAD over the most datasets. This fully demonstrates the advantage of using transformer blocks with position encoding for sequence modeling. ATUAD obtains a lower training time while keeping a slightly higher parameter size than MSCRED, OmniAnomaly, MAD-GAN, and USAD. This illustrates that our ATUAD can effectively accomplish the model training with less calculation. Furthermore, ATUAD achieves a significantly lower parameter size while holding a slightly higher training time than TranAD, which can be explained by the input size of the whole model and its component. On the one hand, the input size of the model affects the parameter size. Specifically, in one iteration, TranAD takes as input a complete batch of time window sequences and a local contextual window, whereas our ATUAD only handles the complete batch of window sequences. As a result, our ATUAD has much fewer parameters (1.35 MB) than TransAD (5.16 MB). On the other hand, the input size of the model's component affects the training time. Concretely, our ATUAD pushes the complete window sequences as the input to its Decoder1 and Decoder2 for aiding temporal attention, while just one local contextual window is fed into TranAD's Window Encoder. The outcome is that our ATUAD consumes more training time while capturing more temporal information. Consequently, the training efficiency and storage overhead of our ATUAD is competitive and acceptable compared to the related baselines.

V. Conclusion

Confront the multivariate time series with complex spatiotemporal correlations and the demand for swift anomaly detection in modern applications, we construct ATUAD, an unsupervised anomaly detection model based on adversarially trained transformers. ATUAD uses a transformer-based encoder–decoder framework to reconstruct data in parallel and applies adversarial training to amplify the deviation of mild anomalies. Besides, a POT-based dynamic threshold mechanism is utilized in ATUAD to determine a threshold for identifying anomalies. Extensive experiments on six public datasets demonstrated that ATUAD outperforms most state-of-the-art baselines on evaluation metrics. Despite ATUAD achieving competitive performance in anomaly detection and training efficiency, centralized anomaly detection on time-series data in a cloud server may lead to privacy breaches, limiting the model's applicability in scenarios with high privacy protection requirements. Consequently, our future work will focus on introducing personalized federated learning and cryptography into our model to make it suitable for handling sensitive heterogeneous data.

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